Welcome to your next lesson: Division modes: / vs // with quotient and remainder.

I'm getting your personalized lesson ready — based on what you've learned, what's next, and what will help most right now.

Let's dive in — you're making real progress.

- ## Imagine you have 36 cookies and each box holds 8 cookies ?

  If you do this in Python:

  ```python
  print(36 / 8)
  ```

  you're asking for the exact math answer. Python will give you a decimal number (a float), not just whole boxes.

  Quick gut check:

  If you run `print(36 / 8)` in Python, which result feels more right?

  A) 4

  B) 4.5

  C) 5

  Just tell me A, B, or C — then we'll build from there step by step.

    - ## I had a more entire-course related question, not sure if you can help me. I was trying to find the course syllabus:
      - how long this class lasts
      - how many weeks
      - how long the entire semester that entails three courses lasts
      - what three courses I have this semester
      - Somewhere I saw that list and I can't find it

    ## Love that you're thinking about the whole path, not just this one lesson ?

    Here's what I can tell you, and then I'll point you to exactly where to look in your account:

1. Where the syllabus lives
   - In the left menu, go to Learn.
   - At the top of that area you should see your current course and, usually, a way to view your program / course list or track.
   - That's where the "three courses this term" list and the sequence normally appear.
2. About timing (general structure)
   - Terms are built around about three courses per term, taken sequentially, not all at once.
   - The exact dates (start/end, how many weeks, which three courses you have this semester) depend on your specific registration, so I can't see those directly from here.

Because I don’t have access to your personal schedule, the best next step is:

- Open Learn → your current program / course,
- Look for something like “Syllabus”, “Course list”, or “Program outline” – that’s where:
  - how long this class lasts
  - how many weeks
  - which three courses this term
  - term/semester length

  are usually shown.

If you open Student Services (left menu, “?” icon) and say “I need my syllabus / course list for this term,” the support team can also pull your exact semester breakdown (with dates) for you.

To help more precisely:

Can you tell me what program name you see at the top of the Learn page (e.g., “AAS in AI Software Engineering”, “Intro to Python Programming”, etc.)? With that, I can describe exactly what you should click next.

Today

Welcome back, Ron! ?

We’re in the Python lesson on division modes — learning the difference between `/` (true division) and `//` (floor division), and how to get quotient + remainder with `//` and `%`.

You’d just thought about:

```python
print(36 / 8)
```

and what that should output.

Quick restart question:

What do you expect this to print in Python?

```python
print(36 / 8)
```

- A) 4
- B) 4.5
- C) 5

Just reply with A, B, or C, and we’ll build from there step by step.

> Somehow we've skipped ahead. I don't recall beginning this lesson or hearing any of that yet but I know that 36 divided by 8 is 4. 5 and I see by way of trying it, that print (36/8) outputs 4.5 and print(36//8) outputs 4

Nice work checking it in Python yourself, Ron — that’s exactly the kind of experimenting I want you to do ?

You've already discovered the key idea of this lesson:

- `36 / 8` → `4.5` → exact division (can be a decimal)
- `36 // 8` → `4` → floor division (whole number, cuts off the decimal)

So:

`/` *keeps the fractional part.*

`//` *drops the fractional part and keeps only the whole-number part.*

Quick visual way to see it:

- 36 cookies, 8 per box
- Exact math: 36 ÷ 8 = 4.5 → "4 and a half boxes" (this is `/`)
- Whole boxes only: you can fill 4 full boxes → that's `//`.

Before we go further (to remainders with `%`), I want to be sure this part is solid:

Does this match how you're thinking about `/` vs `//` now, or is there any part of it that still feels fuzzy?

It makes sense now

Perfect, that's exactly what I needed ?

Now let's layer one small new piece on top: the remainder.

Same cookie story ?

You have 36 cookies, 8 per box.

- `36 // 8` → `4` → 4 full boxes
- But 4 × 8 = 32, so there are 4 cookies left over.

Python has an operator just for that "left over" part: `%`

`36 % 8` *means "what's left over when I divide 36 by 8 using whole numbers?"*

So in Python:

```python
print(36 % 8)
```

will give you `4`.

Does the idea of `%` as "leftover / remainder" make sense so far?

Yes, that makes sense

What do you think? Pick the answer that best matches the cookie story ?

You've seen that `36 // 8` gives `4` and `36 % 8` gives `4`. Together, what do these two numbers represent in a real-world "cookies in boxes" story?

A.

4 full boxes and 4 cookies left over

B.

4.5 boxes exactly, no cookies left over

C.

8 full boxes and 0 cookies left over

D.

A math error in Python

Nice job, Ron — you nailed that one ?

36 // 8 → 4 means 4 full boxes.

36 % 8 → 4 means 4 cookies left over.

So together they perfectly describe the real-world situation:

4 full boxes and 4 extra cookies that don't fit in a full box.

Next tiny step:

Try this in your Python editor (no need to submit, just run it for yourself):

Python
```
print(50 // 9)
print(50 % 9)
```

Then tell me, in plain English, what those two numbers mean in a story like "50 cookies, 9 per box."

There 5 boxes of cookies, 9 per box, with 5 cookies left over (5.5)

Exactly right again ?

For 50 // 9 and 50 % 9:

- 50 // 9 → 5 → 5 full boxes (9 cookies in each box)
- 50 % 9 → 5 → 5 cookies left over

And yes, if you did exact division:

Python
```
print(50 / 9)
```

you'd get about 5.5555... — which matches your "5.5-ish boxes" intuition. cares about the precise math result, // and % care about whole groups + leftovers.

Now I want to show you how to use both at once in Python to see the story clearly.

In your code editor (not here in chat), run this:

Python
```
cookies = 50
per_box = 9

print("Full boxes:", cookies // per_box)
print("Leftover cookies:", cookies % per_box)
```

Run it, look at the output, and then tell me: does it match the story you described?

Yes, it matches

Great ? You've just used // and % together correctly — that's a big part of this lesson.

Let's tie everything together with one more clear, real-world check, then we can decide if you're ready to wrap this lesson.

Think of these three as a team:

- `/` → exact division (can be decimal)
- `//` → how many full groups you can make
- `%` → what's left over after making full groups

Try this new scenario in your head:

You have 29 students and 4 seats per row in a classroom.

- `29 // 4` = ? → full rows
- `29 % 4` = ? → leftover students

What do you think these two numbers would mean in plain English (no need to calculate perfectly, just reason it out)?

> 29 students divided by four rows equals seven rows of students with one student left over

Exactly, Ron — that's spot on ?

For 29 students, 4 seats per row:

- `29 // 4 = 7` → 7 full rows of 4 students
- `29 % 4 = 1` → 1 student left over without a full row

You're clearly seeing how `//` = full groups and `%` = leftovers in real situations. That's exactly the mastery this lesson aims for.

Quick recap check in your own words (one sentence each):

1. What does `/` do when you divide two integers in Python?
2. What does `//` do?
3. What does `%` give you?

Answer briefly, and if those are solid, you're basically done with this lesson ✅

> 1. shows the full integer + quotient
> 2. trunctuates the answer to only show the integer without the quotient
> 3. trunctuates the answer to only show the quotient, ignoring the integer

You're really close, Ron — just need to tweak the meanings a bit ?

Let's line them up clearly with small numbers:

Say we do `11` divided by `4`.

- `11 / 4` → `2.75`
- `/` gives the full exact result, which can be a decimal.
- `11 // 4` → `2`
- `//` gives only the whole-number part (how many full groups of 4 fit in 11). It throws away the decimal.

- 11 % 4 → 3
- % gives only the remainder / leftover (what's left after making full groups):
- 2 groups of 4 = 8
- 11 − 8 = 3 leftover

So:

1. / → exact result (integer + decimal part together, like 2.75)
2. // → whole-number part only (the quotient when you think in full groups), like 2
3. % → remainder only (the leftover), like 3

Can you rewrite your three definitions, super short, like:

1. / → ...
2. // → ...
3. % → ...

using the ideas above: exact result / whole-number part / remainder?

1. / integer + decimal
2. // integer (whole number part) only
3. Remainder only It's honestly what I was saying. I think I just incorrectly called the remainder of the quotient

Yep, you had the right idea — just mixed up the word "quotient" a bit ?

Your cleaned-up versions are perfect:

1. / → integer + decimal (exact result)
2. // → integer (whole-number part) only
3. % → remainder only

That's exactly the mental model I want you to have ?

Last piece to lock in the "teamwork" of // and %:

When you do a // b and a % b:

- a // b → how many full groups of b fit in a
- a % b → what's left over after those full groups

Together, they completely describe the situation (like "7 rows and 1 student left over").

You've hit both lesson goals:

- You can contrast / and // on integers and explain the difference.
- You can use // and % together to get quotient + remainder in real examples.

You've mastered this lesson's core ideas ?

Want to move on to the next lesson, or do you want one more practice scenario with // and % before we wrap this one?